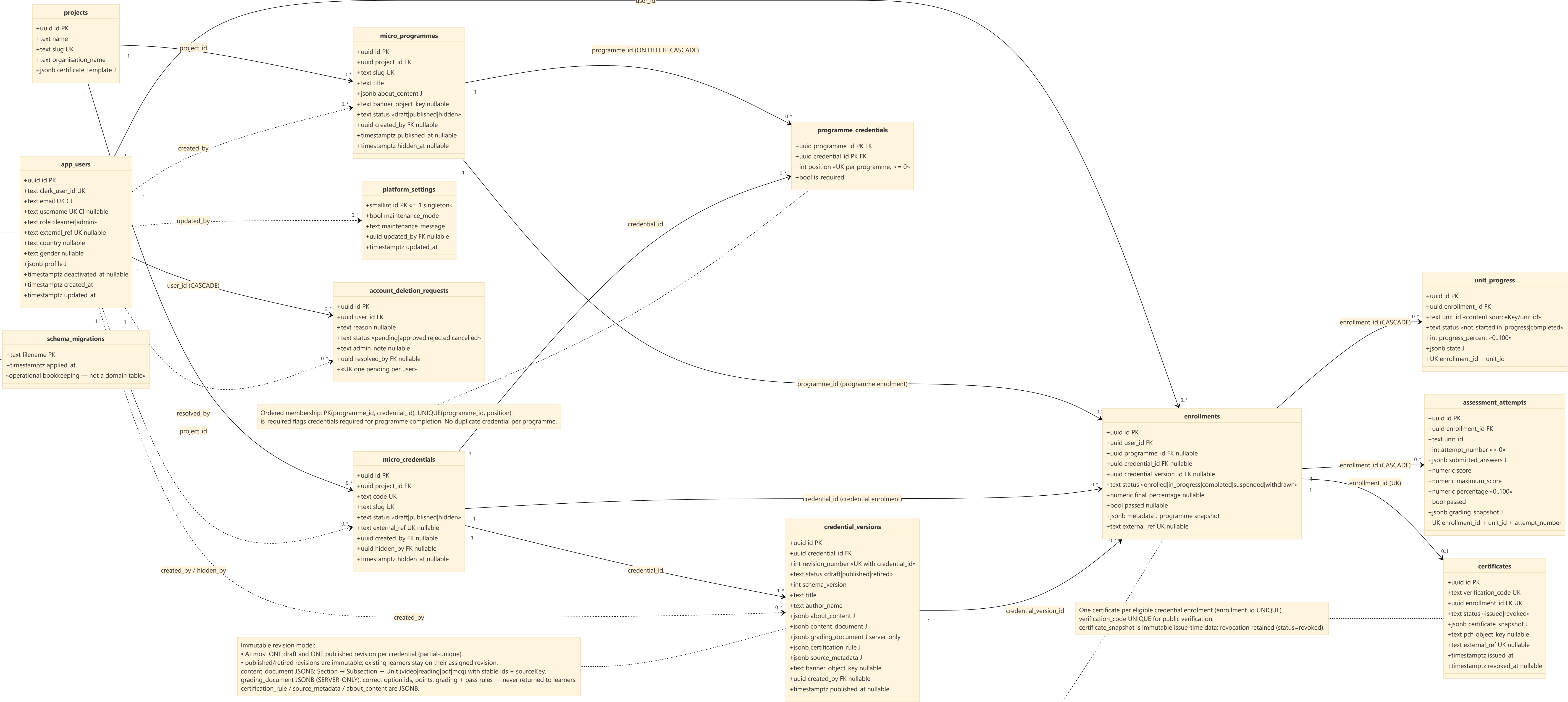


File bytes are NOT stored in PostgreSQL. Banner/PDF/OLX columns hold provider-neutral LOGICAL object keys (e.g. content///, olx//.tar.gz).
Local provider today; switchable to Backblaze B2 with no schema change. No assets table; no absolute paths.

Applied-migration ledger written by scripts/db/migrate.mts. Operational bookkeeping only — separate from the domain tables above.



Ordered membership: PK(programme_id, credential_id), UNIQUE(programme_id, position).
is_required flags credentials required for programme completion. No duplicate credential per programme.

Immutable revision model:
• At most ONE draft and ONE published revision per credential (partial-unique).
• published/retired revisions are immutable; existing learners stay on their assigned revision.
content_document JSONB: Section → Subsection → Unit (video|reading|pdf|mcq) with stable ids + sourceKey.
grading_document JSONB (SERVER-ONLY): correct option ids, points, grading + pass rules — never returned to learners.
certification_rule / source_metadata / about_content are JSONB.

Exactly ONE kind (chk_enrolment_kind):
• Programme enrolment: programme_id set; credential_id & credential_version_id NULL
• Credential enrolment: credential_id + credential_version_id set; programme_id NULL
Partial-unique: one programme enrolment and one credential enrolment per user.
Trigger enforces credential_version_id belongs to credential_id.
metadata JSONB holds the programme registration snapshot (member credential + enrolment ids).

One certificate per eligible credential enrolment (enrollment_id UNIQUE).
verification_code UNIQUE for public verification.
certificate_snapshot is immutable issue-time data; revocation retained (status=revoked).

